

Landslide Risk Assessment Analytics

Problem Statement

India experiences frequent landslides that result in loss of life, infrastructure damage, economic losses, and environmental degradation. Disaster management agencies require a centralized analytics system to identify high-risk regions, understand environmental triggers, evaluate disaster impacts, and improve emergency response planning. This project uses historical landslide data to generate actionable insights that support disaster preparedness and resource allocation.

Objectives

- Identify landslide-prone states and districts.
 - Analyze environmental factors contributing to landslides.
 - Evaluate disaster impacts on people and infrastructure.
 - Assess emergency response efficiency.
 - Analyze resource utilization during rescue operations.
 - Measure recovery and rehabilitation effectiveness.
 - Build interactive dashboards for decision-making.
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Success Metrics (KPIs)

KPI	Description
Total Landslide Events	Total recorded landslide incidents
High-Risk States	States with highest landslide occurrences
High-Risk Districts	Districts most affected by landslides
Total Casualties	Total fatalities and injuries
Economic Loss	Estimated financial damage
Average Response Time	Average emergency response time
Rescue Efficiency	Rescue duration and personnel deployment

Resource Utilization	Equipment and rescue team utilization
Recovery Completion	Percentage of recovery completed
Seasonal Trends	Landslide occurrence by season

Dataset Files

Merged_Landslide_Data.csv

Business Meaning of Dataset Columns

Column	Business Meaning
Event_ID	Unique landslide event identifier
Date	Date of landslide occurrence
State	State where landslide occurred
District	District affected
Latitude	Geographic latitude
Longitude	Geographic longitude
Season	Season during occurrence
Rainfall_mm	Rainfall before landslide
Elevation_m	Elevation above sea level
Slope_Degree	Terrain slope
Soil_Type	Soil classification
NDVI	Vegetation density index
Temperature_C	Temperature during event
Humidity	Humidity level
Distance_to_River_km	Distance from nearest river
Land_Use	Land use classification

Historical_Landslide_Count	Previous landslide occurrences
Landslide_Occurred	Indicates occurrence of landslide
Landslide_Risk	Risk category
Casualties	Number of fatalities
Injured	Number of injured people
Missing_Persons	Missing individuals
Families_Affected	Families impacted
Population_Affected	Population affected
Houses_Damaged	Houses damaged
Roads_Blocked_km	Roads blocked or damaged
Bridges_Damaged	Bridges damaged
Economic_Loss_INR	Estimated economic loss
Cropland_Damaged	Agricultural land damaged
Livestock_Lost	Livestock losses
Response_ID	Emergency response identifier
Response_Time_Min	Response time
Rescue_Duration_Hours	Rescue operation duration
Human_Resources_Deployed	Rescue personnel deployed
Rescue_Teams	Number of rescue teams
NDRF_Teams	National Disaster Response Force teams
SDRF_Teams	State Disaster Response Force teams
Volunteers	Volunteers involved
Ambulances	Ambulances deployed
Helicopters	Helicopters deployed
Excavators	Excavators deployed

JCB_Machines	JCB machines deployed
Cranes	Cranes deployed
Relief_Camps	Relief camps established
Evacuated_People	People evacuated
Aid_Amount_INR	Relief funds distributed
Relief_Materials_Tons	Relief materials supplied
Compensation_Provided	Compensation amount
Recovery_Days	Days required for recovery
Power_Outage_Hours	Duration of power outage
Water_Supply_Disrupted	Water disruption duration
Communication_Disruption	Communication outage duration

Data Dictionary v1

Column	Type	Description	Example
Event_ID	Text	Unique landslide event ID	LS000145
Date	Date	Landslide occurrence date	2024-08-12
State	Text	State name	Kerala
District	Text	District name	Wayanad
Latitude	Decimal	Latitude coordinate	11.685
Longitude	Decimal	Longitude coordinate	76.132
Season	Text	Season of occurrence	Monsoon
Rainfall_mm	Decimal	Rainfall before landslide	285.4
Elevation_m	Integer	Elevation above sea level	980
Slope_Degree	Decimal	Terrain slope	37.6
Soil_Type	Text	Soil classification	Laterite
NDVI	Decimal	Vegetation index	0.72

Temperature_C	Decimal	Temperature	24.5
Humidity	Decimal	Humidity percentage	91.2
Distance_to_River_km	Decimal	Distance from river	1.8
Land_Use	Text	Land use type	Forest
Historical_Landslide_Count	Integer	Previous landslide events	5
Landslide_Occurred	Boolean	Landslide occurrence	Yes
Landslide_Risk	Text	Risk category	High
Casualties	Integer	Fatalities	18
Injured	Integer	Injured people	52
Missing_Persons	Integer	Missing persons	4
Families_Affected	Integer	Families affected	215
Population_Affected	Integer	Population affected	1280
Houses_Damaged	Integer	Houses damaged	162
Roads_Blocked_km	Decimal	Roads blocked	12.5
Bridges_Damaged	Integer	Bridges damaged	2
Economic_Loss_INR	Decimal	Economic loss	35000000
Cropland_Damaged	Decimal	Cropland damaged	65.4
Livestock_Lost	Integer	Livestock lost	34
Response_ID	Text	Response identifier	RESP1024
Response_Time_Min	Integer	Response time	95
Rescue_Duration_Hours	Decimal	Rescue duration	18.5
Human_Resources_Deployed	Integer	Personnel deployed	145
Rescue_Teams	Integer	Rescue teams	8
NDRF_Teams	Integer	NDRF teams	3
SDRF_Teams	Integer	SDRF teams	2
Volunteers	Integer	Volunteers	48

Ambulances	Integer	Ambulances deployed	12
Helicopters	Integer	Helicopters deployed	2
Excavators	Integer	Excavators deployed	6
JCB_Machines	Integer	JCB machines	5
Cranes	Integer	Cranes deployed	2
Relief_Camps	Integer	Relief camps	7
Evacuated_People	Integer	People evacuated	820
Aid_Amount_INR	Decimal	Relief fund	12000000
Relief_Materials_Tons	Decimal	Relief material supplied	145.7
Compensation_Provided	Decimal	Compensation distributed	8500000
Recovery_Days	Integer	Recovery duration	28
Power_Outage_Hours	Integer	Power outage duration	48
Water_Supply_Disrupted	Integer	Water disruption duration	36
Communication_Disruption	Integer	Communication outage duration	18